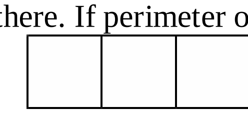
# JMO – 2005

Max Mark – 100 Time – 3 Hrs

Instructions:

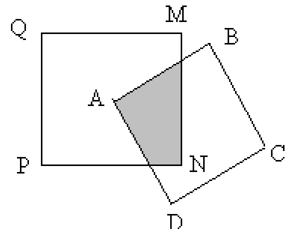
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1. $\left(2\frac{3}{x}\right)\left(y\frac{1}{2}\right)=7\frac{3}{4}$, find the value of x and y.
2. A taken loan of Rs. 800 for 6% rate of interest and B taken loan of Rs. 600 for 10% rate of interest. After what time their interest will be same?
3. A man started from home at 14.30 (by his clock at home) and drove to his village and arrived there when the village clock indicated 15.15. After staying there for 25 minutes, he travel by another road whose distance is $\frac{5}{4}$ times the previous distance but with a speed double of the previous speed and he drove at 16.00(by his clock at home). Find out how much the village clock was fast or late in comparison to his clock at home?
4. In an election from voter list 10% not voted and vote of 60 persons are rejected. In that only two contestants are there. If winner got 47% of voter list and he got 308 more votes than the opposite contestant. Find the numbers in voter list.
5. If 1, 2, 3, 4, …….27, 28, 29, 30 are written on the perimeter of a circle in equal distances. Find the number opposite to 7.
6. Five cards are lying on the table. On each card one English alphabet in one side and one number on other side. Mary said " if one side of card contains a vowel then other side contains odd number
7. Find out smallest 6 digit perfect square number". Nalini picked a card and shows the other side and proved Mary is wrong. So Nalini picked which card from the table?
8. In the figure 3 congruent squares are there. If perimeter of the figure is 80 cm then find its area?

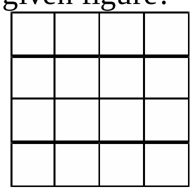


9. A farmer taken 7 kg, 10 kg, 14 kg, 18 kg and 19 kg potatoes in different bags. He sold four bags to two persons such one got double in weight of other. Then find the weight of the potato bag left with him.
10. In Mathematics 78%, in algebra 83%, in geometry 88% and in trigonometry 97% students passed. Then at least what % of students passed in all the subjects?
11. Find out two numbers such that their HCF is 4 and their LCM is 24. Also find their sum.
12. Find the largest of the following numbers. $2^{175}$, $3^{125}$, $4^{100}$, $5^{75}$, $6^{50}$ .
13. Write in how many ways we can represent 70 as a sum of two prime numbers?
14. To travel from home to school it takes 30 minutes to me and 40 minutes to my brother. If my brother starts 5 minutes earlier to me, then after what time I will meet my brother?
15. Jyoti lies on Monday, Tuesday, Wednesday and tells the truth on all other days. Similary Kamala lies on Thursday, Friday, Saturday and tells the truth on all other days. Teir conversations are like:
    Jyoti: I lie on Saturday.
    Kamala: I will lie on tomorrow.
    Then on what days they made the conversation?

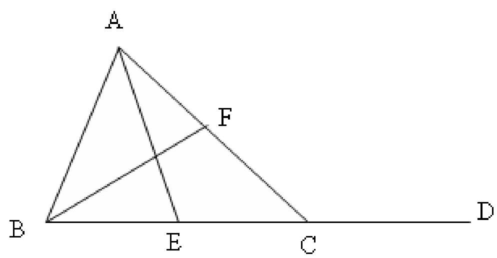
16. PQMN and ABCD two equal squares drawn like the figure such that centre of the square PQMN is A and the side AB intersect 1/3 rd of side MN. Then shaded area is what portion of the area of PQMN?




17. The pages of a book are numbered consecutively: 1, 2, 3, 4, …., 11, 12, …. And so on. No pages are missing. If in page numbers the digit 3 occurs 99 times what is the number of the last page?
18. How many 2X 2 squares are there in the given figure?



19. After simplification of $4^{2001}$ we got a number. What are the last two digits of that number?
20. In the figure $m\angle CAE = 30^0, m\angle BAE = 70^0$. Find $m\angle ACD = ?$




21. If 8mn9 is a perfect square then find $m^2 + n^2 = ?$
22. A person sold one cow at a profit of 16%. If he bought that cow at 24% less and sold Rs 86 less, then he will get a profit of 30%. Find cost price of the cow.
23. After ringing 5 clocks at a time, then after 3, 5, 7, 8 and 10 seconds again they start ringing. Then in one week how many times 5 clocks rang at the same time?
24. A had Rs. 9 and B had Rs. 4.20. After lost a game, A gave some money to B, it seen that A had $\frac{5}{6}$th of B. Then how much B got from A?
25. A monkey climbs a pole 3 metre in each second and after 3 seconds he slips by 4 meter. If height of the pole is 40 meter then how much time monkey will take to climb at the top of the pole?

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